

KHALIL IBRAHIM

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Portfolio: sirballoon.github.io/DevBook

PROFESSIONAL SUMMARY

Embedded systems and software engineer with hands-on experience in C/C++, Python, and JavaScript. Proven ability to develop IoT devices, desktop applications, and database systems. Strong background in hardware-software integration, ESP32 microcontroller programming, and full-stack development. Seeking to leverage technical expertise and problem-solving skills in embedded systems engineering.

TECHNICAL SKILLS

- **Embedded Systems:** C, C++, ESP32, Arduino, Hardware-Software Integration, Bluetooth Low Energy, E-Paper Displays
- **Programming Languages:** Python, C, C++, JavaScript, TypeScript, SQL
- **Frameworks & Tools:** React, Node.js, PyQt, Git, REST APIs, Web3
- **Databases:** MySQL, MongoDB, SQL scripting, Database optimization
- **Development Practices:** MVC Architecture, Multithreading, Agile Development, Version Control, Problem-Solving

TECHNICAL PROJECTS

E-Paper Keychain | C++, ESP32-S3, E-Paper Display, 3D Printing

- Designed and developed a compact IoT device featuring an ESP32-S3-Zero microcontroller and e-paper display
- Implemented low-power display refresh algorithms optimized for battery life in embedded systems
- Created custom 3D-printed enclosure integrating hardware components with mechanical design constraints

Bluetooth Barcode Scanner Retrofit | C++, ESP32, Bluetooth, Hardware Integration

- Reverse-engineered USB barcode scanner protocol to enable wireless Bluetooth connectivity via ESP32
- Developed firmware for real-time data transmission between salvaged hardware and Bluetooth host devices
- Demonstrated proficiency in embedded systems debugging and hardware-software interface design

Neurolyzer - Neurophysiological Analysis Platform | Python, PyQt, MVC Architecture, Multithreading

- Led full-stack development of desktop application analyzing neurophysiological data for neurodegenerative disease research
- Architected MVC design pattern with Qt threading system for responsive real-time signal processing
- Deployed cross-platform application (Windows/Mac) validated against PhD-level research papers
- Collaborated in agile four-person team, managing version control and sprint cycles

Decentralized Data Sharing Platform - OpenD/I Hackathon | JavaScript, TypeScript, Node.js, REST APIs, Web3

- Developed backend services using Node.js and REST APIs for cloud-native web application
- Integrated Web3 technologies for blockchain-based decentralized data sharing functionality
- Rapidly adapted to new technology stack in fast-paced hackathon environment, demonstrating learning agility

PROFESSIONAL EXPERIENCE

Software Developer Intern | SATHI (Remote) | Sept 2024 - Nov 2024

- Engineered SQL scripts to clean and standardize medical datasets according to SNOMED CT healthcare standards
- Maintained relational database integrity through schema modifications and data validation in MySQL environment
- Gained domain expertise in medical terminology structures and database-driven healthcare applications

Database Technician & Pharmacy Assistant | Cincotta Chemist, Merrylands | Feb 2022 - Present

- Maintained and updated RoboPharma automated dispensing database ensuring accurate medication inventory management

- Troubleshoot automation system issues in collaboration with pharmacists, interfacing with Fred Dispense software
- Developed adaptability across multiple roles including technical support, customer service, and inventory systems

EDUCATION

Bachelor of Computer Science | Western Sydney University | *Expected Graduation: 2026*

- Major: Artificial Intelligence | Minor: Cloud Computing
- Relevant Coursework: Embedded Systems, Data Structures, Algorithms, Computer Architecture, Operating Systems

CERTIFICATIONS & AWARDS

Available upon request - Google Drive portfolio with technical certifications and project documentation